

# Does Costco Entry Lower Competitor Fuel Prices in Australia?

A Synthetic-Control Study for the ACCC

MGT159 Group Project — Costco Australia

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## Question and motivation

We advise the **Australian Competition and Consumer Commission (ACCC)**, Australia’s national competition regulator. The ACCC publishes regular fuel-price monitoring reports and has flagged ongoing concerns about market concentration in Australian fuel retailing. One of the ACCC’s main tools for improving competition in concentrated retail markets is reducing entry barriers for discount entrants. Whether such entrants actually discipline incumbent pricing is an open empirical question.

We ask:

**Does the entry of a Costco gas station into an Australian local market cause nearby competitor stations to lower their retail fuel prices?**

This is a causal question. We answer it with **synthetic control**, fit separately to each of the four Costco fuel stations in Australia with enough pre- and post-opening price history to support estimation: Perth Airport (WA, Feb 2020), Casuarina (WA, Nov 2022), Lake Macquarie (NSW, May 2022), and Coomera (QLD, May 2023). The pre-committed plan (identifying assumptions, sample construction, robustness checks) is in `deliverables/plan_of_attack_section_{1,2,3}.pdf`.

Table 1: State fuel-price registries used in this analysis.

| Registry      | State | Coverage            | Raw records | Unique stations |
|---------------|-------|---------------------|-------------|-----------------|
| NSW FuelCheck | NSW   | Dec 2016 – Jan 2026 | 4,401,971   | 4,813           |
| QLD FPRS      | QLD   | Dec 2018 – Dec 2025 | 4,970,650   | 1,883           |
| WA FuelWatch  | WA    | Jan 2018 – Apr 2026 | 8,047,153   | 1,423           |

Table 2: Australian Costco fuel stations by treatment status.

| Costco           | State | Treatment date | Pre months | Post months | Status                    |
|------------------|-------|----------------|------------|-------------|---------------------------|
| Coomera          | QLD   | May 2023       | 51         | 31          | Treated                   |
| Casuarina        | WA    | Nov 2022       | 58         | 42          | Treated                   |
| Perth Airport    | WA    | Feb 2020       | 26         | 74          | Treated                   |
| Lake Macquarie   | NSW   | May 2022       | 58         | 35          | Treated                   |
| Marsden Park     | NSW   | Jul 2017       | 11         | 105         | Excluded (pre too short)  |
| Auburn           | NSW   | Aug 2025       | 107        | 8           | Excluded (post too short) |
| Canberra Airport | ACT   | Nov 2023       | 0          | 29          | Excluded (no pre)         |
| Casula           | NSW   | Dec 2016       | 4          | 113         | Excluded (pre-registry)   |
| North Lakes      | QLD   | Jan 2019       | 1          | 89          | Excluded (pre-registry)   |
| Ipswich          | QLD   | Mar 2019       | 3          | 87          | Excluded (pre-registry)   |

## 1 Raw Data Description

### 1.1 Data sources

We combine three publicly accessible Australian state fuel-price registries with a hand-collected list of Costco fuel-station opening dates. The registries are **NSW FuelCheck** (data.nsw.gov.au), the **Queensland Fuel Price Reporting Scheme** (data.qld.gov.au), and **WA FuelWatch** (data.wa.gov.au). The three registries together yield approximately **17.4 million station-day price records across 6,084 unique stations between December 2016 and April 2026**. Each registry mandates public price reporting by every retail fuel station in the state.

NSW FuelCheck and QLD FPRS record price-change events (a row per station  $\times$  fuel type  $\times$  price-update timestamp); WA FuelWatch publishes daily snapshots (a row per station  $\times$  fuel type  $\times$  calendar day). All three are harmonised in `scripts/synthetic_control_input/build_sc_inputs.py` to a common station  $\times$  month panel of mean unleaded price in cents per litre. Detailed field-level schemas appear in `deliverables/plan_of_attack_section_1.pdf` §1.1–1.3.

### 1.2 Treated Costco fuel stations

Ten Costco fuel stations operate (or recently opened) in Australia. Four satisfy the synthetic-control inclusion rule ( $\geq 24$  pre-treatment months and  $\geq 12$  post-treatment months in the relevant state’s registry) and form the treated panel; the remaining six are excluded from formal estimation but appear here as descriptive context.

The four treated Costcos give us **staggered treatment timing across three states** (Feb 2020, May 2022, Nov 2022, May 2023), which is the variation the synthetic-control design exploits.

### 1.3 The analysis panel

The build pipeline cleans the raw registries (adds station coordinates, keeps only regular unleaded, drops implausible quotes, aggregates monthly). The analysis panel that comes out has two pieces:

Table 3: Analysis-ready panel sizes after sample-construction filters.

| Component                         | Rows   | Units                                              |
|-----------------------------------|--------|----------------------------------------------------|
| Treated 5 km rings $\times$ month | 375    | 4 treated Costcos                                  |
| Donor postcodes $\times$ month    | 17,578 | 196 donor postcodes (NSW = 104, QLD = 56, WA = 36) |
| Combined analysis panel           | 17,953 | treated + donor                                    |

The sample-construction funnel (raw rows  $\rightarrow$  aggregated cells  $\rightarrow$  final panel) is reported in §2.5 as documentary text, since reconstructing it inside this .qmd would require pulling the full 1.86 GB of raw API archives.

#### 1.4 Distribution of monthly mean unleaded price

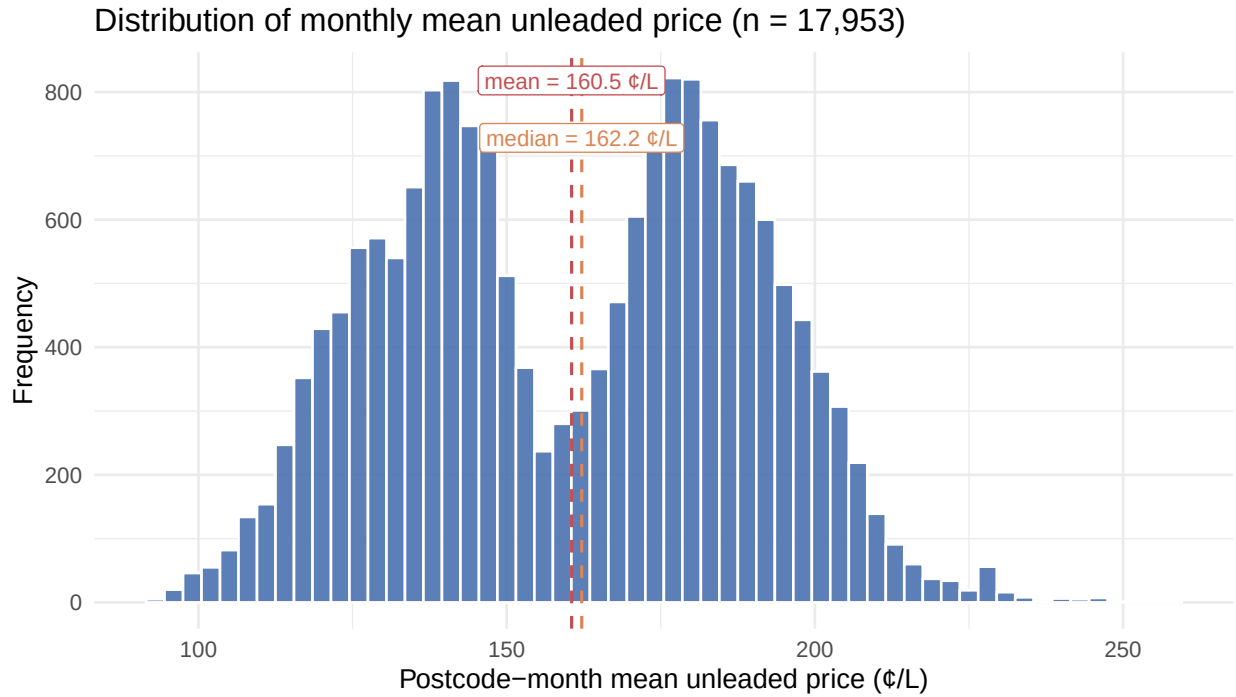


Figure 1: Distribution of monthly mean unleaded price across the combined SC analysis panel (treated 5 km rings + donor postcodes). The distribution is approximately bimodal: the lower mode reflects the 2018–2020 price regime; the higher mode reflects the post-2022 elevated price era following the Russia/Ukraine energy shock.

## 1.5 Monthly median unleaded price by state

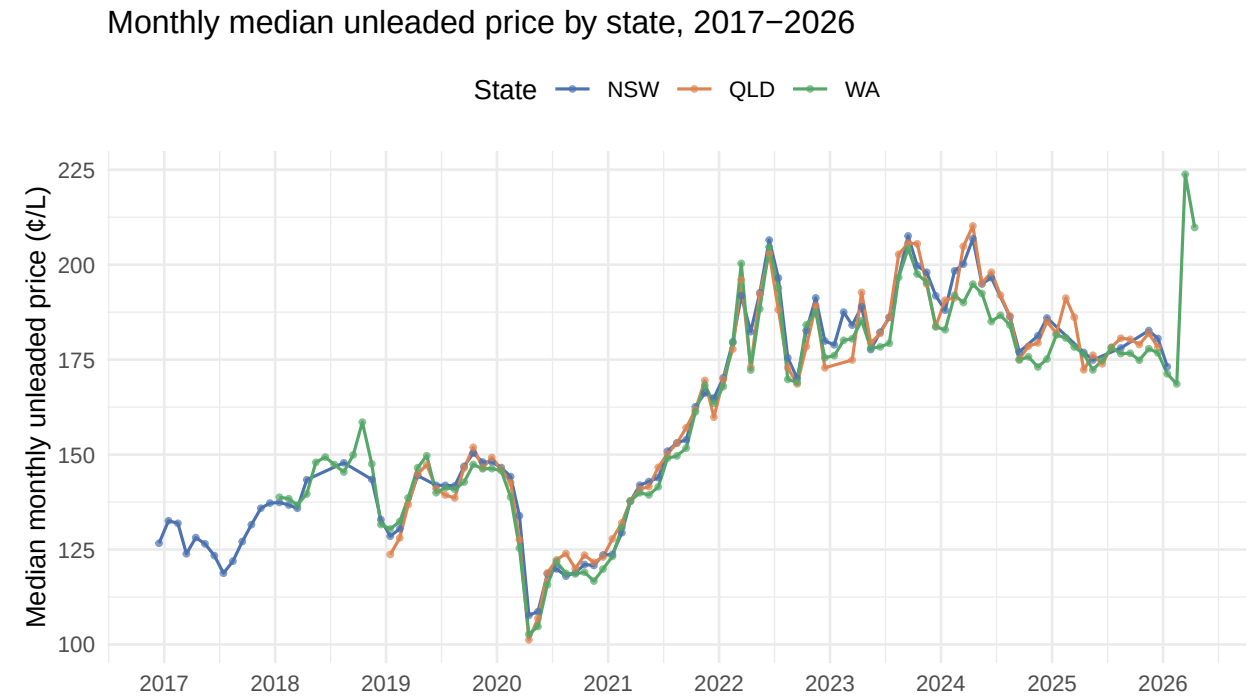


Figure 2: Median of postcode-monthly mean unleaded prices over time, by state. All three states track each other closely: prices fall in 2018–2020 (peaking with the COVID demand collapse in early 2020), rise sharply through 2022 (Russia/Ukraine energy shock), dip during the Mar–Sep 2022 federal fuel-excise cut, and remain elevated through 2026. State-level co-movement supports our use of within-state donor pools to absorb common national shocks.

Table 4: Summary statistics on the SC analysis panel.

| Variable                                       | n      | Mean  | SD    | Min  | p25   | Median | p75   | Max     |
|------------------------------------------------|--------|-------|-------|------|-------|--------|-------|---------|
| Monthly mean unleaded price (¢/L)              | 17,953 | 160.5 | 28.4  | 92.0 | 137.3 | 162.2  | 183.6 | 259.0   |
| Donor-postcode distance to nearest Costco (km) | 196    | 218.7 | 326.6 | 20.2 | 36.7  | 104.7  | 228.4 | 2,152.1 |

## 1.6 Treated 5 km rings versus state medians



Figure 3: Treated 5 km ring mean unleaded price (colored solid) versus the state median across all reporting postcodes (gray dashed) for each of the four treated Costcos. The vertical red dashed line marks the validated treatment date. Shaded gray: the COVID demand collapse (Mar–Sep 2020); dotted vertical: Russia’s invasion of Ukraine (24 Feb 2022); pale-blue shading: the AU federal fuel-excise cut (30 Mar – 28 Sep 2022). The state-median overlay is a descriptive level reference, not the synthetic-control counterfactual.

## 1.7 Summary statistics

The mean unleaded price row matches the combined treated + donor sample. The distance row is computed at the donor-postcode level (one row per postcode); the plan-of-attack Section 1 reports the analogous statistic at the station level ( $n = 5,980$ , mean  $\approx 205$  km), so the numbers differ in denominator but agree in shape: most non-Costco stations sit very far from any Costco.

## 1.8 Data quality and known limitations

- **Price-unit harmonisation.** NSW and WA prices are recorded in cents/L; QLD records prices in tenths of cents (e.g.  $1899 = 189.9$  ¢/L). The build pipeline applies the appropriate scaling per registry so the analysis panel is uniform in cents/L.
- **Missing coordinates in historical NSW and WA files.** Both registries publish historical files without lat/lng; we resolve this with a one-time pull from each state’s live API, joined to historical records by (normalized name, postcode). Stations that closed or rebranded before the live-API snapshot fail the join and drop out (52.3% of NSW historical stations, 47.3% of WA). The loss falls almost entirely on the donor pool; all four treated Costcos join cleanly.
- **NSW Costco rename in May 2022.** Every NSW Costco station was renamed in the registry from e.g. *Costco Marsden Park* to *Costco Marsden Park (Members only)* on 2 May 2022. The build pipeline strips the suffix before matching.
- **Sentinel-price filter.** Quotes outside 50–500 ¢/L are dropped before aggregation; this removes obvious out-of-stock placeholders like 9999 or 0.01. After the filter, no postcode-month in the analysis panel falls outside 92–259 ¢/L (well inside the plausible Australian range).
- **Casuarina sparse ring.** Casuarina’s 5 km ring contains only about three contributing stations on average; this limitation was flagged in advance. Robustness Check 3 (Casuarina at 10 km) probes whether this noise drives the headline.
- **Perth Airport opens four weeks before COVID.** Its early post-treatment months are dominated by pandemic-driven price moves; Robustness Check 4 (drop Mar–Dec 2020) probes whether this confounds the headline.



Table 5: Pre- and post-treatment coverage per treated Costco.

| Costco         | State | Treatment date | Pre months | Post months |
|----------------|-------|----------------|------------|-------------|
| Coomera        | QLD   | May 2023       | 51         | 31          |
| Casuarina      | WA    | Nov 2022       | 58         | 42          |
| Perth Airport  | WA    | Feb 2020       | 26         | 74          |
| Lake Macquarie | NSW   | May 2022       | 58         | 35          |

## 2 Sample Construction

Each subsection below walks through one sample-construction choice, in the same order as plan-of-attack §2(a)–(g). The choices here describe the 5 km treated ring / 20 km donor buffer used in §3; §4 tests alternate geometries.

### 2.1 Sample window

The pre-period for each treated Costco runs from the start of its state’s price registry (December 2016 for NSW, January 2018 for WA, December 2018 for QLD) through the day before that Costco’s validated treatment date (§2.4). The post-period runs from the treatment date through April 2026, the most recent registry pull. Per-Costco coverage:

Synthetic control fits weights to match the treated unit’s full pre-period trajectory. More pre-period months tighten the fit; unlike DID, longer pre-periods don’t risk breaking parallel trends in this design. Same-state donors absorb common shocks (oil-price changes, fleet turnover, federal excise adjustments) by construction. We use as much pre-period as the data allow.

**COVID.** The 2020 demand collapse and recovery is the only structural break large enough to warrant explicit handling. Three of the four treated Costcos opened well after the COVID disruption (Lake Macquarie May 2022, Casuarina November 2022, Coomera May 2023); for those three the COVID period sits entirely inside the pre-period and is matched by the synthetic by construction. **Perth Airport (treated 19 February 2020) opens approximately four weeks before Australia’s COVID demand collapse begins; this is the most fragile case in the analysis.** Its pre-period is entirely pre-COVID, but its early post-treatment months are dominated by pandemic-driven price movements rather than competitive response to Costco’s entry. The synthetic counterfactual is built from WA donor postcodes that experienced the same pandemic shock, so the post-treatment gap nets out COVID provided the pre-period weights track the treated trajectory through January 2020. §4 Robustness Check 4 re-estimates Perth Airport with March–December 2020 dropped from both panels.

### 2.2 Other sample restrictions

**Treated unit.** For each treated Costco, the treated unit is the *local market* defined as all non-Costco retail stations within 5 km of the Costco’s coordinates, summarized each month as the simple mean of those stations’ monthly mean unleaded prices. Costco’s own price quotes are excluded from the treated mean. The research question concerns competitor response; Costco’s own pricing is outside scope.

**Donor pool.** Each donor unit is a 4-digit Australian postcode, summarized the same way (monthly mean of non-Costco unleaded price across all stations within the postcode). Postcodes are the natural geographic unit of all three state registries and are consistent across NSW, QLD, and WA. A postcode is excluded from the donor pool if any of its stations lie within 20 km of any Costco fuel station (treated, excluded, or otherwise), so donor postcodes are unambiguously outside any plausible Costco competitive footprint.

**Asymmetric radii (5 km treated, 20 km donor buffer).** The economics literature on retail fuel-price competition (Houde 2012; Eckert 2013) finds price effects concentrated within roughly 1–5 miles. 5 km (~3.1

Table 6: Threshold-and-defense summary for the SC sample.

| Threshold              | Value              | Justification                                                                   | Robustness check (§4)                             |
|------------------------|--------------------|---------------------------------------------------------------------------------|---------------------------------------------------|
| Treated radius         | at most 5 km       | Conservative end of IO-literature retail fuel-price spillover range (1–5 mi)    | Re-run at 3 km and 8 km                           |
| Donor buffer           | > 20 km            | 4× the treated radius; keeps donors fully outside any Costco’s competitive zone | Re-run at 15 km and 30 km                         |
| Donor stations / month | at least 3         | Below 3, monthly mean has high sampling variance                                | Re-run at at least 2 and at least 5               |
| Donor coverage         | at least 24 months | Minimum pre-period length for stable SC weight fitting                          | Re-run at at least 36 months                      |
| Costco pre-period      | at least 24 months | Same logic, applied to treated                                                  | Not relaxed (would re-introduce excluded Costcos) |
| Costco post-period     | at least 12 months | Minimum to estimate a sustained-effect gap                                      | Not relaxed                                       |
| Plausible price range  | 50–500 ¢/L         | Sentinel-value floor and ceiling; AU retail unleaded lives in 80–280 ¢/L        | None; not an analytic threshold                   |

mi) sits at the conservative end of that range: small enough to capture genuinely competing stations, large enough to include most of the local market. The 20 km donor buffer is intentionally a 4× multiple, providing a wide exclusion donut so that no donor postcode is partly affected by a Costco that isn’t the one we’re studying. The asymmetry biases the design *against* finding an effect: a narrow treated definition shrinks the treated population to the most exposed stations, and the wide buffer shrinks the donor pool to truly untreated postcodes.

**Outcome variable.** Regular unleaded: `FuelCode` in {U91, E10, P91} for NSW; `Fuel_Type` in {Unleaded 91, e10} for QLD; `PRODUCT_DESCRIPTION` = ULP for WA. Unleaded is the highest-volume retail product in Australia, comparable across the three state schemes, and Costco’s headline competitive product.

**Donor-pool measurement-quality filters.** A postcode is additionally excluded if it averages fewer than 3 unique stations per month (monthly means based on 1–2 stations are too noisy to serve as credible counterfactual building blocks) or has fewer than 24 months of any observed coverage (weights cannot be reliably fit on a short pre-period).

**Treated-Costco inclusion rule.** A Costco fuel station is treated as a synthetic-control unit if and only if its registry history contains both **at least 24 pre-treatment months** and **at least 12 post-treatment months** in the relevant state’s price registry. Four of Australia’s ten Costco fuel stations satisfy both conditions; six do not and appear as descriptive context only (§1.2).

## 2.3 Dependent variable integrity

**Construction.** The dependent variable is the monthly mean unleaded retail price (cents per litre) at the treated 5 km ring or the donor postcode; see §2.5 for the two-stage construction. Figure 1 shows the post-aggregation distribution ( $n = 17,953$  postcode-months, range 92–259 ¢/L, approximately bimodal across the pre-2022 / post-2022 price-regime divide).

**Sentinel-value filter.** Individual price quotes outside 50–500 ¢/L are dropped before aggregation. This filter targets registry artifacts (out-of-stock placeholder values like 9999 or 0.01); legitimate tail observations pass through. It is not a winsorisation step.

**Missing months.** A postcode-month or treated ring-month with zero contributing station observations is left as missing (NA), never imputed as zero, never carried forward. Synthetic-control fitting handles missing months by ignoring them in the pre-period objective. §1’s data-quality notes report that the largest run of missing months in any treated panel is three (Lake Macquarie), well within tolerance.

Table 7: Validated treatment dates: registry first-appearance cross-checked against fuel-station-specific corroboration. Perth Airport is the only revision.

| Costco         | Registry first-appearance | Fuel-station-specific corroboration                                                    | Delta | Treatment date used            |
|----------------|---------------------------|----------------------------------------------------------------------------------------|-------|--------------------------------|
| Coomera        | 23 May 2023               | 23 May 2023 (Retail World; explicit fuel-station opening)                              | 0 d   | 23 May 2023 (registry)         |
| Casuarina      | 11 Nov 2022               | None; trade press only reports the warehouse opening (7 Dec 2022)                      | n/a   | 11 Nov 2022 (registry)         |
| Perth Airport  | 11 Apr 2020               | 19 Feb 2020 (ACAPMAG: petrol station selling at \$1.18/L; warehouse opens 19 Mar 2020) | -52 d | <b>19 Feb 2020 (announced)</b> |
| Lake Macquarie | 2 May 2022                | None; no fuel-station-specific opening date in public sources                          | n/a   | 2 May 2022 (registry)          |

**No winsorisation in the main specification.** Two-stage monthly aggregation smooths individual-quote outliers: an extreme single quote at one station gets averaged with the rest of that station’s month, and then with the rest of the ring or postcode. If a §4 robustness check exposes a donor-postcode month driving an estimate, we winsorise the analysis panel at the 99th percentile and report both specifications side by side.

## 2.4 Independent variable integrity

**Treatment indicator.** A binary post-opening flag is constructed per treated Costco by merging the hand-collected treatment-date table onto the (Costco  $\times$  month) panel: 1 from the treatment month forward, 0 before.

**Treatment-date validation.** Registry first-appearance can lag the true fuel-station opening if a station begins selling fuel before appearing in the state’s reporting feed. Each of the four treated Costcos’ registry-first-appearance dates was cross-checked against Costco Australia corporate announcements, trade press, and news coverage. The corroborating source must specifically reference the *fuel station* rather than the warehouse; Costco Australia warehouses and fuel stations regularly open weeks apart. If a fuel-station-specific announced opening date differs from registry first-appearance by more than 30 days, the announced date supersedes.

Three of the four registry first-appearance dates either match the fuel-station-specific corroborated date exactly (Coomera) or lack any fuel-station-specific external date for comparison (Casuarina, Lake Macquarie); these retain the registry date. **Perth Airport is the only revision:** the ACAPMAG trade-press report explicitly distinguishes the petrol station (open 19 Feb 2020, bowser price quoted) from the warehouse (opening one month later on 19 Mar 2020), placing the fuel-station opening 52 days before the first FuelWatch record.

**Station coordinates and distance-to-Costco.** Distance is computed as the great-circle distance from each station’s coordinates to the nearest Costco fuel station’s coordinates. NSW and WA historical registry files do not carry lat/lng, so coordinates are joined from one-time pulls of the live FuelCheck and FuelWatch APIs by (normalized name, postcode); QLD’s historical files include coordinates inline. Stations that fail the coordinate join are dropped from analysis. Without a known location, they cannot be classified into a 5 km ring or a donor postcode. Per-registry join failure rates are in §2.6.

## 2.5 Aggregation

**Why monthly.** Three considerations point at month as the right level of aggregation. First, NSW and QLD are event-based registries (rows are price-change events) while WA is a daily-snapshot registry; month is the lowest common frequency at which all three speak the same language without having to invent ‘daily’

observations for the event-based feeds. Second, the research question is about *sustained* competitive response to Costco entry, which plays out over weeks-to-months rather than days, so daily resolution would add noise without identification. Third, monthly means across multiple stations in a ring or postcode smooth single-quote outliers without requiring any winsorisation step.

**Two-stage aggregation.** Within each station  $\times$  month, we compute a single mean unleaded price across all of that station’s price observations in the month. We then take a simple (unweighted) mean across stations to produce the (ring  $\times$  month) treated panel and the (postcode  $\times$  month) donor panel. The two-stage structure prevents stations that change prices more often (and therefore appear more times in the event-based NSW/QLD feeds) from contributing disproportionately to the cell mean.

**Carry-forward step (NSW and QLD only).** For NSW and QLD, where rows are price-change events rather than daily snapshots, we construct each station’s daily price series by carrying its last-observed price forward to subsequent days within the month until the next observed change. This is the standard approach for converting price-event data to a daily series and then to monthly means. WA’s daily-snapshot data passes through unchanged.

## 2.6 Data merging

Three sources are joined into a single analysis panel:

- **State registries** (NSW FuelCheck, QLD FPRS, WA FuelWatch) provide the price observations.
- **Live-API station snapshots** (NSW, WA) and inline coordinates (QLD) provide station latitude / longitude, joined to historical registry rows by (normalized station name, postcode).
- **Hand-collected Costco metadata** (`data/catalogs/usable_costcos.csv`) provides each Costco’s lat/lng and validated treatment date, joined to treated panels by Costco name.

**Expected losses in the merge.** Two sources of leakage. First, stations that closed, rebranded, or were re-numbered before the April 2026 live-API pull are absent from that snapshot, so historical rows that mention them fail the coordinate join. The snapshot misses **2,376 of 4,543 NSW historical stations (52.3%)** and **673 of 1,423 WA historical stations (47.3%)**; QLD is unaffected because its historical files carry coordinates inline. The loss falls almost entirely on the donor pool (all four treated Costcos join cleanly), and 196 donor postcodes survive the downstream geographic and quality filters. Second, station-name typos and mid-panel renames occasionally break the join; the build pipeline addresses these programmatically where it can (the May 2022 NSW Costco “Members only” rename, for example, is stripped before matching).

## 2.7 Filters and transformations: the funnel from raw to analysis-ready

The funnel below traces the exact row and cell counts from the raw registry pulls through to the four analysis-ready files in `data/sc_inputs/`. Row counts in the upper block give way to (station  $\times$  month) cells, then (postcode  $\times$  month) cells, then postcode counts, then Costco counts as the panel is aggregated and filtered. **Source:** [scripts/synthetic\\_control\\_input/build\\_sc\\_inputs.py](#). Bolded rows are the milestone outputs.

**Output.** Four CSVs in `data/sc_inputs/`: `treated_units.csv` (375 Costco-month rows), `donor_pool.csv` (17,578 postcode-month rows), `treated_metadata.csv` (4 rows), and `donor_metadata.csv` (196 rows). These four files, uploaded to the Dropbox folder referenced in the setup chunk, form the complete input to §3’s synthetic-control analysis.

Table 8: Sample-construction funnel from raw registry rows to the analysis-ready panels in `data/sc_inputs/`.

| Step                                                   | NSW        | QLD        | WA        | Justification                                                              |
|--------------------------------------------------------|------------|------------|-----------|----------------------------------------------------------------------------|
| Raw registry pull (rows)                               | 4,401,971  | 4,970,650  | 8,047,153 | Source data                                                                |
| Filter to regular unleaded (rows)                      | 1,982,507  | 997,680    | 2,059,755 | Question is about retail unleaded; premium/diesel are different markets    |
| Drop sentinel prices 50–500 ¢/L (rows)                 | 1,982,506  | 995,145    | 2,059,755 | Out-of-stock placeholder removal; nearly nonbinding                        |
| Join station coordinates (rows)                        | 1,247,728  | 995,145    | 1,289,912 | Required for radius assignment; NSW/WA from live-API snapshot              |
| Aggregate to (station × month) cells                   | 156,864    | 70,564     | 68,332    | Carry-forward for NSW/QLD events; passthrough for WA                       |
| Aggregate to (postcode × month) cells                  | 42,307     | 18,186     | 18,089    | Donor candidate panel (561 / 245 / 198 postcodes)                          |
| Drop postcodes at most 20 km from any Costco           | -231 pcds  | -51 pcds   | -80 pcds  | Donors must be outside any Costco’s competitive footprint                  |
| Drop postcodes < 3 stations / mo avg                   | -226 pcds  | -138 pcds  | -82 pcds  | Monthly mean too noisy below 3 stations                                    |
| Drop postcodes < 24 months observed                    | -0 pcds    | -0 pcds    | -0 pcds   | Coverage was good; no postcode failed this gate                            |
| <b>Final donor pool (postcodes)</b>                    | <b>104</b> | <b>56</b>  | <b>36</b> | <b>196 total × ~90 months = 17,578 (postcode × month) cells</b>            |
| Costco eligibility (at least 24 pre, at least 12 post) | -4 Costcos | -2 Costcos | 0         | Inclusion rule applied to all 10 Australian Costco fuel stations           |
| <b>Final treated panel (Costcos)</b>                   | <b>1</b>   | <b>1</b>   | <b>2</b>  | <b>4 Costcos × 5 km ring × ~80–100 months = 375 (Costco × month) cells</b> |

## 3 Synthetic-Control Analysis

### 3.1 Research design

**Question type.** Causal. The identification strategy is **synthetic control**, fit separately for each of the four treated Costcos.

**Why synthetic control over difference-in-differences.** With four treated Costcos and 196 candidate donor postcodes, a treated-versus-control mean comparison is too thin: a single anomalous Costco (e.g. Casuarina, with only ~3 contributing stations) would dominate the average. Synthetic control builds a weighted counterfactual for each treated unit individually. This is the design the  $n = 4$  case calls for.

**Source of variation.** Costco fuel-station entry into a local market isn’t driven by what’s happening to local fuel prices. Site selection is driven by Costco warehouse logistics, land availability, and Costco’s national rollout schedule. None of these respond to the local fuel-price trajectory at the chosen site. The variation we exploit is the staggered timing of the four openings (Feb 2020, May 2022, Nov 2022, May 2023) across three states.

**Identifying assumptions.** Each is paired with the figure or check that defends it:

- *Pre-period fit captures unobserved time-varying confounders.* Verified visually in Figure 4 (pre-period overlap between treated and synthetic), quantified in Table 10 (pre-period tracking error per Costco), and tested by the 12-month holdout in §4 Robustness Check 5.
- *Donors stay clean.* No donor postcode is partly affected by a Costco that isn’t the one we’re studying. Defended by the 20 km donor buffer (4× the treated radius), pre-committed in §2.2, and explicitly tested by §4 Robustness Check 7 (spatial placebo on the 5–20 km donut).

- *No anticipation.* Competitor stations do not preemptively cut prices ahead of Costco’s announced opening. Tested by §4 Robustness Check 1 (placebo in time,  $-12$  months).
- *Range coverage.* The treated ring’s pre-period price level can be expressed as a weighted average of donor postcodes. Verified by the donor-weight bars in Figure 5 (no negative weights, no extreme single-donor concentration).

**What the weights do.** One synthetic-control fit per treated Costco. For each Costco the donor pool is restricted to **same-state postcodes only** so the synthetic absorbs state-specific shocks (state excise rates, regional refining capacity, intra-state shipping costs) by construction. Weights are chosen to minimize the average gap between the treated 5 km ring’s monthly mean unleaded price and the weighted donor mean across the pre-period.

**Method commitment.** Standard synthetic control via R’s `tidysynth` package. **Predictors:** yearly means of pre-period unleaded price (no auxiliary covariates; no postcode-level demographic panel justifies them here). **Pre-period objective:** minimize the average gap between treated and synthetic on monthly mean unleaded price (`optimization_window = pre_period` in `tidysynth::generate_weights`). **Inference:** treat each donor postcode as a fake treated unit and measure the resulting fake effect; report the post/pre tracking-error ratio and a 95% CI on the post-treatment gap. All fitting runs through `_sc_helpers.R::fit_one_costco()`.

## 3.2 Fitting the four synthetic controls

The chunk above fits four `tidysynth` models (one per treated Costco), each with `generate_placebos = TRUE` so the donor-permutation distribution is computed in the same pass. Caching is enabled (`cache: true`) so subsequent knits re-use the fits.

## 3.3 Synthetic-control trajectories per Costco

Four stacked panels of treated vs synthetic trajectories. Solid colored = actual treated 5 km ring; dashed black = synthetic counterfactual; vertical dotted red = validated treatment date (§2.4); shaded gray band = 95% donor-permutation CI on the synthetic. The CI comes from re-fitting the synthetic on each donor postcode as if it were treated, then taking the 2.5–97.5 percentile of those placebo gaps at each time point. A wider band at a given month means more uncertainty at that point; a treated curve that falls outside the band is evidence the gap is larger than donor-side variability alone would produce.

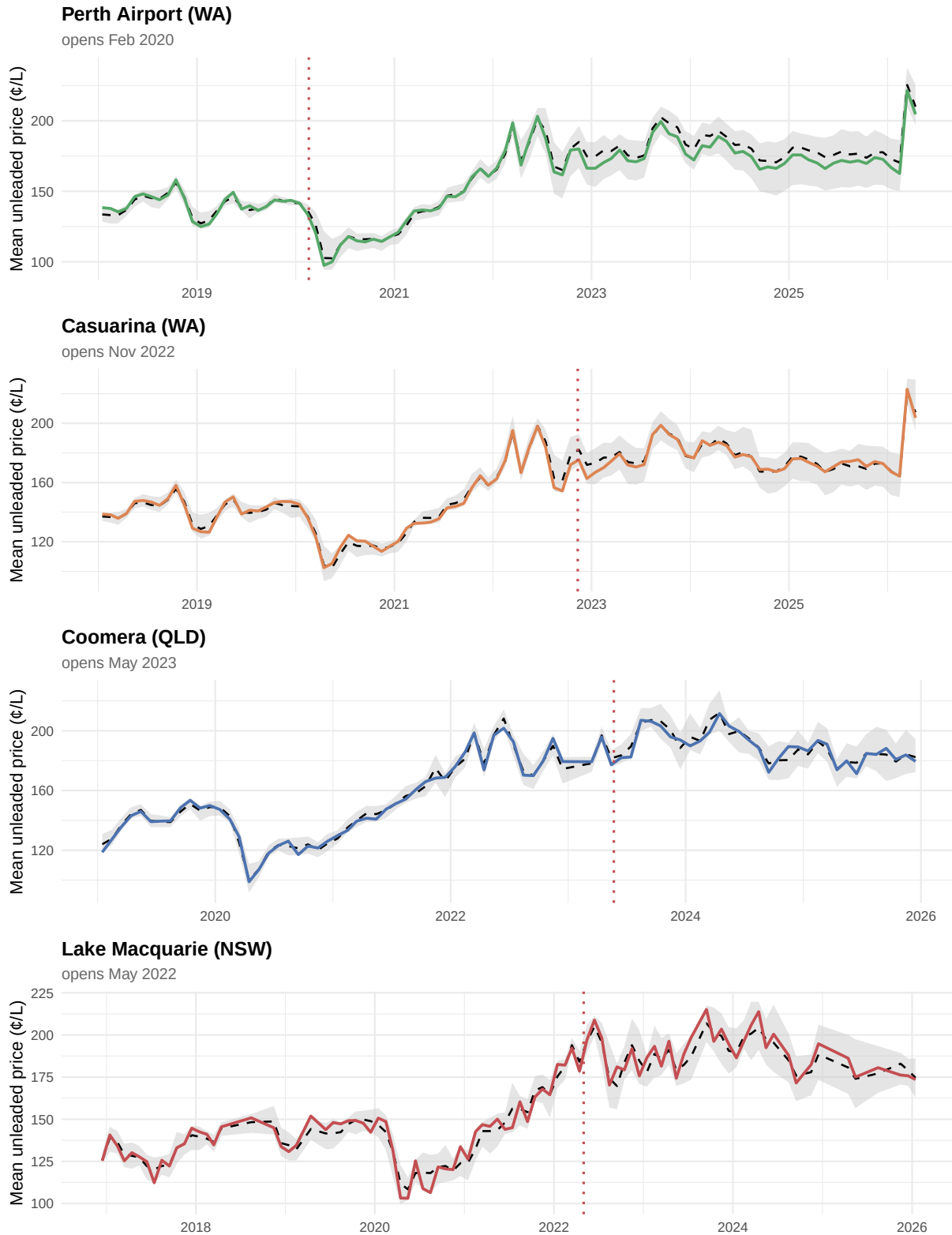


Figure 4: Synthetic-control trajectories per treated Costco, stacked for full-width time comparison. Solid colored = treated 5 km ring (actual); dashed black = synthetic counterfactual; vertical dotted red = validated Costco opening; shaded gray = 95% donor-permutation CI on the synthetic.

**Reading the panels.** Perth Airport (top) shows the treated 5 km ring sitting below its synthetic across much of the post-2020 window, the clearest visual gap among the four. Casuarina and Coomera have treated and synthetic tracking closely through both pre- and post-periods, consistent with little detectable competitor response in their 5 km rings. Lake Macquarie’s treated line runs slightly above its synthetic in the post-period (about +1.5 ¢/L); §4 RC7 shows the surrounding 5–20 km donut going strongly negative, so the wrong-sign Lake Macquarie headline is best read as a 5 km artifact.

### 3.4 Donor weights per treated Costco

For each Costco we report the postcodes whose synthetic-control weight exceeds 0.01, sorted in descending order. A counterfactual leaning on one or two donors is fragile; one drawn from a balanced mix of geographically sensible donors is credible. Geographic sanity-check: Coomera should lean on Gold Coast / Brisbane postcodes, Lake Macquarie on Newcastle / Central Coast, Perth Airport and Casuarina on Perth metro postcodes.

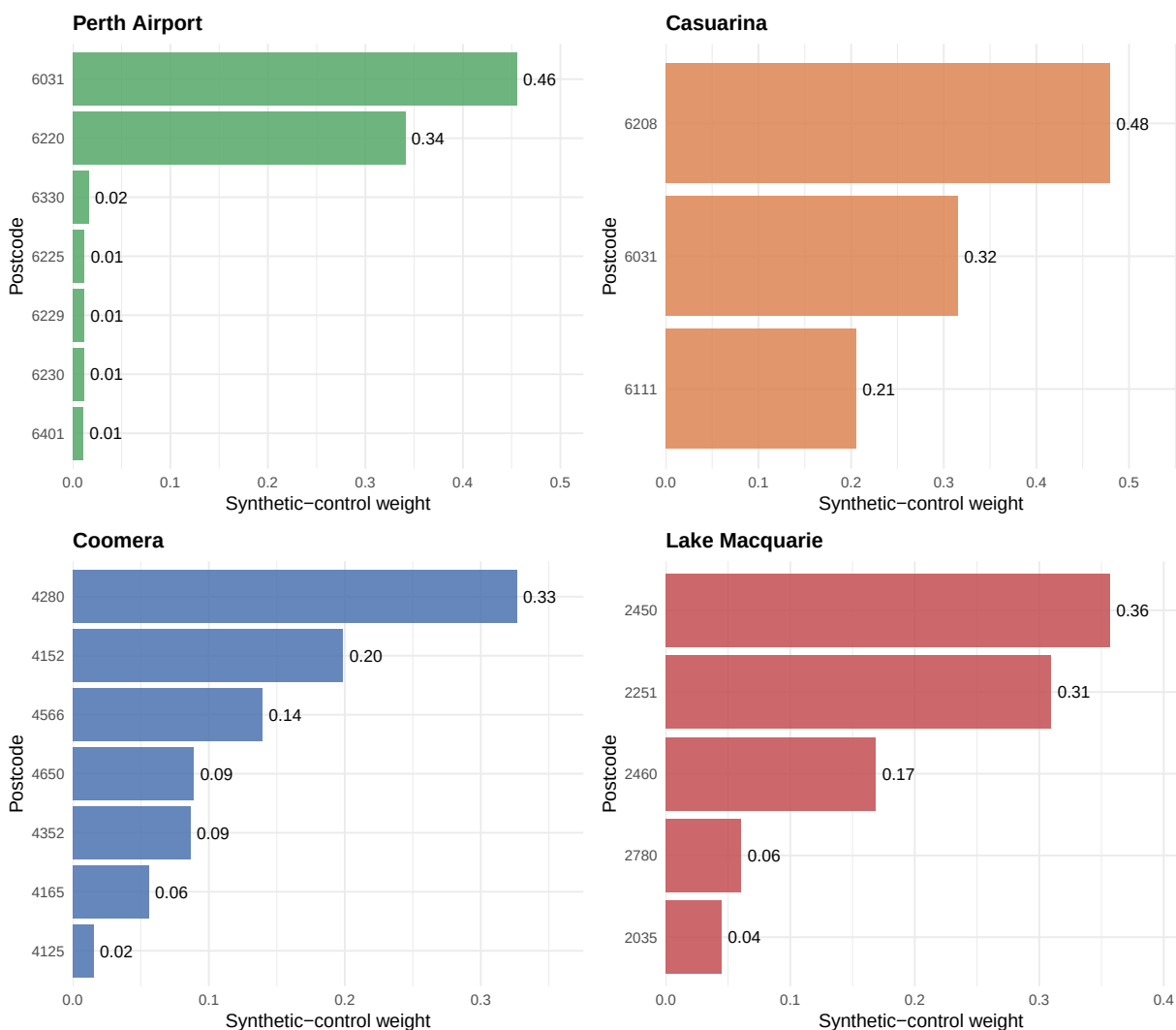


Figure 5: Donor-postcode weights per treated Costco (weights > 0.01, sorted descending). A balanced mix across 5-7+ donors with no single donor exceeding ~0.5 is the credibility signal; concentration on 1-2 donors would flag a fragile counterfactual.



### 3.5 Aggregate event study

Figure 4 shows each Costco separately. Figure 6 stacks them on a common timeline to ask how big the effect is on average, how fast it appears, and how long it lasts. The window is  $-24$  to  $+24$  relative months, the largest range over which all four Costcos contribute (limited by Perth Airport's 26 pre-months and Coomera's 31 post-months).

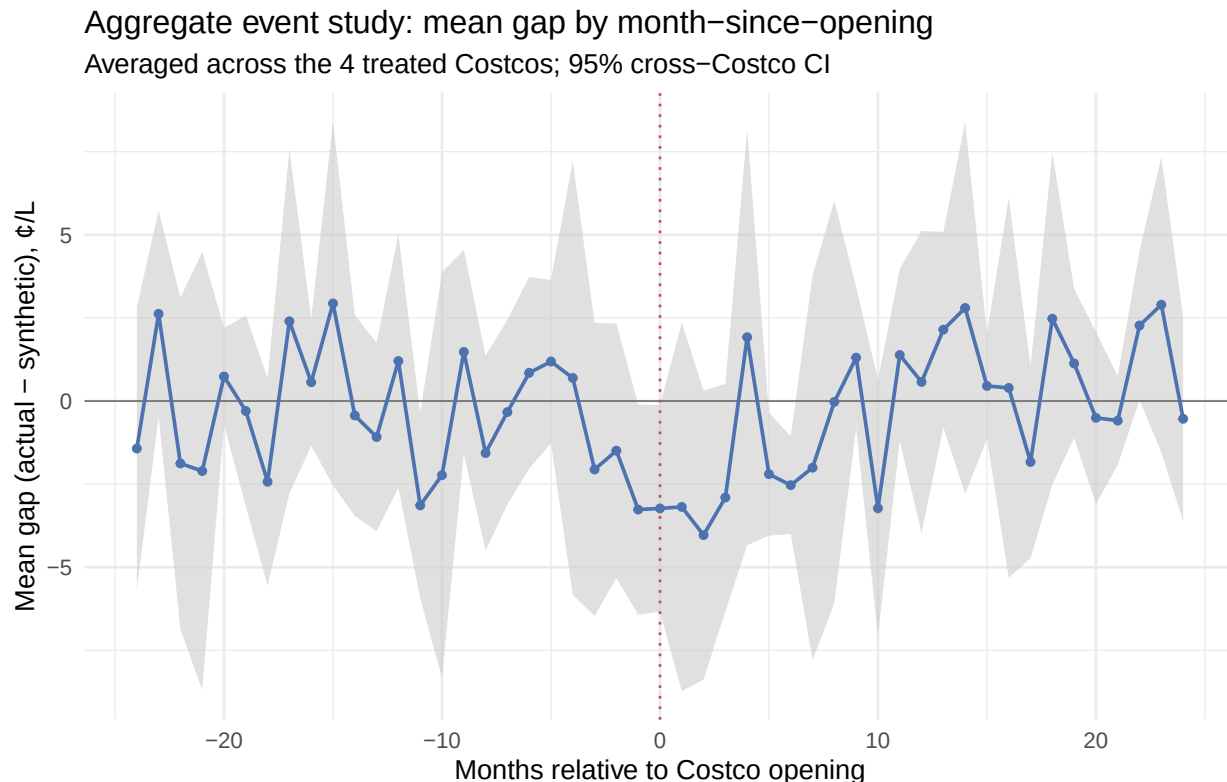


Figure 6: Aggregate event study. X-axis: months relative to Costco opening. Y-axis: mean of (actual - synthetic) gap across the four treated Costcos. Shaded gray: 95% CI from cross-Costco standard error. Vertical dotted red: treatment (month 0).

**Reading the figure.** The mean gap fluctuates within roughly  $\pm 3$  c/L across the  $-24$  to  $+24$  window, with the cross-Costco 95% CI band always crossing zero. The aggregate is essentially null on its own: Perth Airport's negative gap is offset by Lake Macquarie's positive one and Coomera's near-zero gap in the four-Costco mean. The per-Costco picture in Table 9 is what carries the substantive answer.

### 3.6 Per-Costco effect-size summary

The **post/pre tracking-error ratio** compares the synthetic's miss after Costco entry to its miss before. A ratio of 4–5 or higher is the standard bar for a real effect: the post-period gap is several times larger than the pre-period noise. The **95% CI** comes from treating each donor postcode as a fake treated unit and taking the 2.5–97.5 percentile of those placebo gaps. A CI that excludes zero is evidence against the no-effect null.

**Reading the rows.** Perth Airport is the only Costco that clears the MSPE-ratio bar (4.31 against the 4–5 threshold) and the only one whose 95% CI lies entirely below zero. Coomera's MSPE ratio of 2.08 sits below the bar and its CI straddles zero. Lake Macquarie's CI is entirely above zero, flagging a wrong-sign result that \$5.1 reads as a 5 km artifact (RC7's surrounding donut goes strongly negative). Casuarina's

Table 9: Per-Costco effect-size summary. Mean post-treatment gap = mean of (actual - synthetic) across the post-treatment window. Post/pre MSPE ratio compares post-period mean-squared prediction error to pre-period; values of 4-5+ are the standard threshold for evidence of a real effect.

| Costco         | State | Pre mean<br>(¢/L) | Post gap<br>(¢/L) | % red. | MSPE<br>ratio | 95% CI         | Post<br>mo. |
|----------------|-------|-------------------|-------------------|--------|---------------|----------------|-------------|
| Casuarina      | WA    | 143.04            | -0.91             | -0.63  | 1.00          | [-0.91, -0.90] | 42          |
| Coomera        | QLD   | 151.27            | -0.60             | -0.40  | 2.08          | [-0.56, 0.85]  | 31          |
| Lake Macquarie | NSW   | 140.23            | 1.52              | 1.08   | 1.40          | [0.45, 1.49]   | 35          |
| Perth Airport  | WA    | 140.74            | -3.40             | -2.41  | 4.31          | [-3.35, -1.37] | 74          |

Table 10: Pre-period fit quality per treated Costco. Lower tracking error means a tighter pre-period match; a tracking error below 1% of the pre-period mean is typically considered a credible fit. Top-3 donor weight share reflects how concentrated the synthetic is on its strongest donors.

| Costco         | Pre months | Pre tracking error<br>(¢/L) | Tracking error as % of<br>pre mean | Top-3 donor weight<br>share |
|----------------|------------|-----------------------------|------------------------------------|-----------------------------|
| Casuarina      | 58         | 2.537                       | 1.77                               | 1.00                        |
| Coomera        | 51         | 2.649                       | 1.75                               | 0.66                        |
| Lake Macquarie | 58         | 4.860                       | 3.47                               | 0.83                        |
| Perth Airport  | 26         | 2.218                       | 1.58                               | 0.81                        |

CI of [-0.91, -0.90] is suspiciously narrow: with only ~3 contributing stations in its 5 km ring, the donor-permutation distribution collapses because placebo fits return near-identical post-treatment means via the predictor fallback. The headline Casuarina row is statistically degenerate, so we treat it as uninformative and lean on RC3’s 10 km read in §5.1.

### 3.7 Pre-period fit quality

Casuarina’s pre-period tracking error is visibly larger than the others’ because its 5 km ring contains only ~3 contributing stations on average (§1’s documented limitation).

Table 11: RC1 placebo-in-time: mean (actual - synthetic) gap over the 12 months between the placebo date and the real treatment date. A gap near zero — and a 95% CI that includes zero — is the pass criterion.

| Costco         | State | Placebo date | Real date | Mean placebo gap (¢/L) | 95% CI         |
|----------------|-------|--------------|-----------|------------------------|----------------|
| Casuarina      | WA    | Nov 2021     | Nov 2022  | -4.83                  | [-4.69, 0.67]  |
| Coomera        | QLD   | May 2022     | May 2023  | 2.51                   | [-0.15, 2.05]  |
| Lake Macquarie | NSW   | May 2021     | May 2022  | -3.43                  | [-3.35, -0.24] |
| Perth Airport  | WA    | Feb 2019     | Feb 2020  | -1.96                  | [-2.09, -0.35] |

## 4 Robustness Checks

The biggest sources of fragility in the headline analysis are (i) the 5 km / 20 km geographic geometry, (ii) the COVID period coinciding with Perth Airport’s opening, (iii) the noise in Casuarina’s sparse 5 km ring, (iv) the standard synthetic-control concern that pre-period fit can be gamed by over-flexible donor selection, and (v) the lack of formal uncertainty quantification on a four-Costco point estimate. The seven pre-committed checks below directly probe these concerns. Checks are numbered to match plan-of-attack §3(c). Checks 2, 3, and 7 use alternate-radius SC inputs built by [scripts/synthetic\\_control\\_input/build\\_sc\\_inputs\\_alt\\_radii.py](#) and pulled from the same Dropbox folder as the headline inputs.

### 4.1 Placebo in time (RC1)

**Concern.** No anticipation: competitor stations do not pre-emptively cut prices ahead of Costco’s announced opening. If a ‘ban-style effect’ appears in the placebo window before Costco actually opens, the headline gap is contaminated by pre-existing trends or anticipation.

**Procedure.** For each treated Costco, backdate treatment by 12 months to define a placebo opening date. Refit synthetic control on the truncated panel (all months strictly before the real treatment date), so no real post-treatment observations contaminate the placebo window. Compute the mean placebo gap over the 12 months from the placebo date to the real treatment date, and report the donor-permutation 95% CI.

**Interpretation.** The pass criterion is a placebo gap near zero with a 95% CI that includes zero. This means donor-side variability is wider than the placebo gap, so the headline post-treatment gap cannot be attributed to a pre-existing downward trend or to anticipation in the 12 months before Costco’s actual opening.

### 4.2 Alternative radii (RC2)

**Concern.** The 5 km treated radius and 20 km donor buffer are defensible (Houde 2012, Eckert 2013 place price effects within 1–5 miles; the 4× buffer keeps donors fully outside the competitive zone) but discretionary. An effect that exists at 5 km but vanishes at 3 km or 8 km would be fragile. This is the most important sensitivity test because the radius geometry *is* the design.

**Procedure.** Re-build SC inputs at (3 km treated, 15 km donor buffer) and (8 km treated, 30 km donor buffer). Refit each Costco at each geometry and report per-Costco gaps in a sensitivity table.

**Interpretation.** Treat the 5/20 km row as the reference. For each Costco we check (a) that the post-treatment gap stays negative at both alt radii and (b) that magnitudes do not collapse. A sign flip at any alt radius (or a magnitude reduction of more than ~50%) flags that Costco’s headline estimate as geometry-sensitive.

Table 12: RC2 sensitivity to radius geometry. Per-Costco gaps at the headline 5/20 km specification and at the two alternate geometries (3/15 km, 8/30 km). The headline conclusion is robust if sign and approximate magnitude survive at both alt radii.

| Specification           | Costco         | Pre mean (¢/L) | Post gap (¢/L) | % red. | MSPE ratio | 95% CI         |
|-------------------------|----------------|----------------|----------------|--------|------------|----------------|
| 5 km / 20 km (headline) | Casuarina      | 143.04         | -0.91          | -0.63  | 1.00       | [-0.91, -0.90] |
| 5 km / 20 km (headline) | Coomera        | 151.27         | -0.60          | -0.40  | 2.08       | [-0.56, 0.85]  |
| 5 km / 20 km (headline) | Lake Macquarie | 140.23         | 1.52           | 1.08   | 1.40       | [0.45, 1.49]   |
| 5 km / 20 km (headline) | Perth Airport  | 140.74         | -3.40          | -2.41  | 4.31       | [-3.35, -1.37] |
| 3 km / 15 km            | Coomera        | 151.71         | 0.06           | 0.04   | 2.40       | [0.08, 0.82]   |
| 3 km / 15 km            | Lake Macquarie | 140.42         | 4.25           | 3.03   | 2.48       | [0.68, 4.16]   |
| 3 km / 15 km            | Perth Airport  | 140.46         | -4.87          | -3.47  | 14.75      | [-4.77, -0.67] |
| 8 km / 30 km            | Casuarina      | 141.18         | -4.50          | -3.19  | 2.47       | [-4.40, -0.78] |
| 8 km / 30 km            | Coomera        | 151.29         | 3.89           | 2.57   | 4.38       | [0.10, 3.79]   |
| 8 km / 30 km            | Lake Macquarie | 138.54         | -1.64          | -1.18  | 1.47       | [-1.58, 0.82]  |
| 8 km / 30 km            | Perth Airport  | 140.29         | -0.99          | -0.71  | 0.66       | [-0.99, -0.96] |

Table 13: RC3 Casuarina at the headline 5 km radius vs the 10 km robustness radius. Pre-period tracking error is expected to fall under the wider ring (more contributing stations smooth idiosyncratic noise); the question is whether the post-treatment gap moves correspondingly.

| Spec            | Pre mean (¢/L) | Post gap (¢/L) | % red. | MSPE ratio | Pre tracking err (¢/L) | 95% CI         |
|-----------------|----------------|----------------|--------|------------|------------------------|----------------|
| 5 km (headline) | 143.04         | -0.91          | -0.63  | 1.00       | 2.54                   | [-0.91, -0.90] |
| 10 km (RC3)     | 143.54         | -4.49          | -3.13  | 3.19       | 2.58                   | [-4.43, -2.15] |

### 4.3 Casuarina at 10 km treated radius (RC3)

**Concern.** Casuarina’s 5 km ring contains only ~3 contributing stations on average (documented in §1’s data-quality notes). A wider ring captures more signal at the cost of including some less-exposed stations. If the headline Casuarina estimate flips or shrinks substantially at 10 km, the per-Costco estimate may be driven by individual-station noise rather than competitive response.

**Procedure.** Re-fit Casuarina specifically at a 10 km treated radius (donor buffer remains 20 km). Compare gap and pre-period tracking error to the 5 km baseline.

**Interpretation.** Pre-period tracking error should drop at 10 km because the wider ring includes more contributing stations and the monthly mean is less noisy. If the post-treatment gap is similar at both radii, the headline Casuarina estimate is not driven by individual-station noise. A large swing (especially a sign flip) would mean the 5 km point estimate is unreliable and should be de-emphasized in the headline narrative.

### 4.4 Perth Airport with the COVID period removed (RC4)

**Concern.** Perth Airport opens 19 February 2020, approximately four weeks before Australia’s COVID demand collapse. Early post-treatment months are dominated by pandemic-driven price movements rather than competitive response. COVID-era compression of fuel prices may suppress the actual - synthetic gap; the COVID-removed estimate may be larger in magnitude than the full-window gap.

**Procedure.** Re-fit Perth Airport with March–December 2020 dropped from both the treated and donor panels. Compare the resulting gap to the headline Perth Airport gap.

Table 14: RC4 Perth Airport with the COVID period removed, compared to the headline full-window estimate. A larger gap in the COVID-removed specification indicates that pandemic compression of fuel prices suppressed the headline estimate.

| Specification              | Pre mean (¢/L) | Post gap (¢/L) | % red. | MSPE ratio | 95% CI         | Post mo. |
|----------------------------|----------------|----------------|--------|------------|----------------|----------|
| Headline (full window)     | 140.74         | -3.40          | -2.41  | 4.31       | [-3.35, -1.37] | 74       |
| RC4 (Mar-Dec 2020 dropped) | 140.74         | -3.65          | -2.59  | 4.75       | [-3.60, -1.55] | 64       |

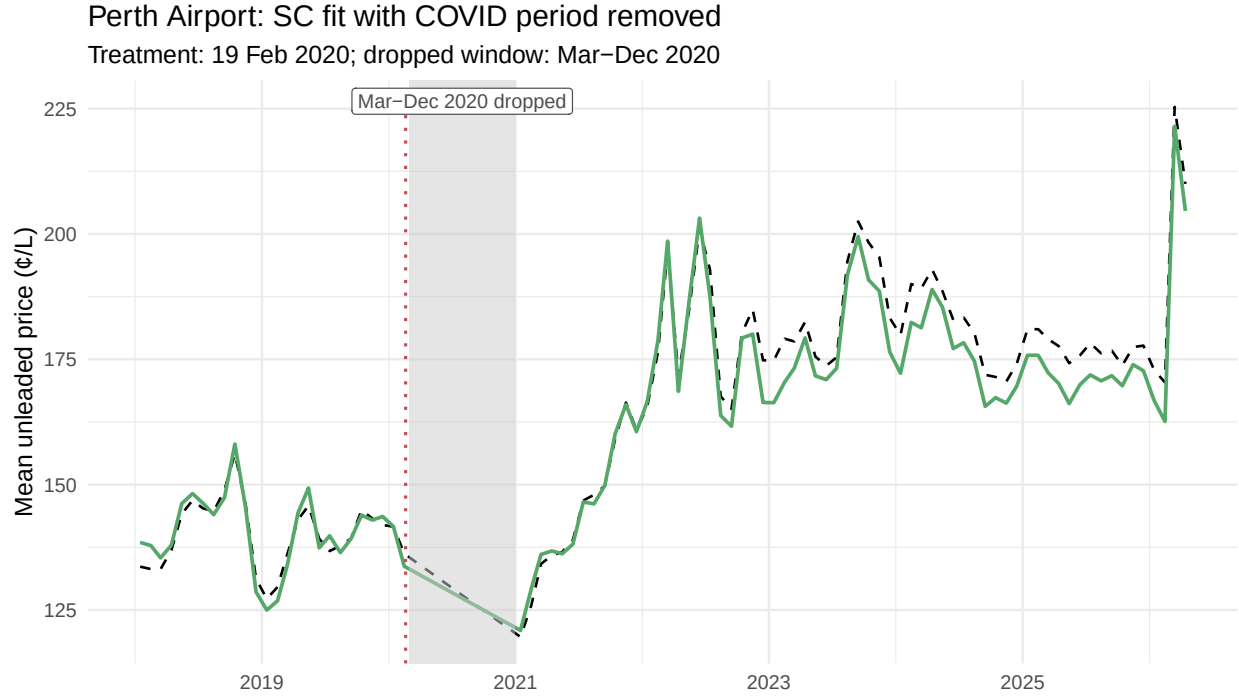


Figure 7: Perth Airport synthetic-control trajectory with March-December 2020 dropped from both treated and donor panels. Solid green = treated 5 km ring; dashed black = synthetic counterfactual; vertical dotted red = validated treatment date (19 Feb 2020). The gap in early 2020 is the dropped window.

**Interpretation.** If the COVID-removed gap is *more negative* than the headline, COVID-era price compression suppressed the actual - synthetic divergence and the headline understates the true effect at Perth Airport. If the two estimates are close, the headline already nets out the pandemic by leaning on within-state donors that experienced the same shock.

#### 4.5 LOOCV donor selection with 12-month holdout (RC5)

**Concern.** Pre-period fit can be gamed by over-flexible donor selection: the synthetic ‘memorises’ the training series without generalising. The honest test is an out-of-sample holdout: did the weights learn structural similarity, or did they memorise noise?

**Procedure.** For each Costco, refit the synthetic on the pre-period *minus the last 12 months* (the training window). The untouched 12-month holdout sits between the training window and the real treatment date. Without looking at any post-treatment data, score whether the synthetic continues to track the treated series across the holdout: holdout RMSPE comparable to training RMSPE means the fit generalises; a much larger holdout RMSPE means the fit was memorising.

Table 15: RC5 12-month holdout. Training window = pre-period minus last 12 months. Holdout = untouched last 12 months of pre-period. Post = post-treatment window. A holdout-to-training RMSPE ratio near 1 means the synthetic generalises out-of-sample; a ratio much larger than 1 means the headline pre-period fit was memorising training noise.

| Costco         | Train months | Train RMSPE (¢/L) | Holdout RMSPE (¢/L) | Post RMSPE (¢/L) | Holdout / Train |
|----------------|--------------|-------------------|---------------------|------------------|-----------------|
| Casuarina      | 46           | 3.09              | 5.99                | 4.68             | 1.94            |
| Coomera        | 40           | 3.01              | 4.24                | 4.05             | 1.41            |
| Lake Macquarie | 46           | 4.97              | 6.09                | 5.25             | 1.22            |
| Perth Airport  | 13           | 1.19              | 2.41                | 6.55             | 2.03            |

The plan-of-attack also describes leave-one-out cross-validation over donor-pool sizes  $N \in \{5, 10, 20, 50, \text{full}\}$ . We skip this step. tidysynth has no native  $N$ -restriction, and writing one would reintroduce the researcher discretion the holdout test is meant to remove. The holdout diagnostic runs on the full donor pool.

**Interpretation.** The holdout-to-training ratio is the headline diagnostic. Values near 1 mean the synthetic learned a structural match; values substantially  $> 1$  (say,  $> 3$ -4) mean the pre-period fit reported in Table 10 may have been generous about generalisation. The post-RMSPE column is shown for orientation: it should be much larger than holdout-RMSPE if the post-treatment gap is a genuine treatment effect rather than fit drift.

## 4.6 95% CIs via donor-permutation inference (RC6)

**Concern.** Point estimates without uncertainty bands can make a modest effect look airtight. Casuarina’s documented noise problem (Table 10) is the case most at risk of overclaiming precision.

**Procedure.** Treat each donor postcode as a placebo treated unit, refit synthetic control on the remaining donors, and compute the resulting placebo gap. Across all donor placebos, take the empirical 2.5–97.5 percentile of post-treatment gaps as a 95% CI on the real treatment effect.

**Implementation.** This is the same donor-permutation distribution that powers the shaded bands in Figure 4 and the 95% CI column in Table 9. Both were computed as part of the §3 fits (`generate_placebos = TRUE` in `_sc_helpers.R`). A CI that excludes zero is evidence against the no-effect null; Casuarina-specific results will be read transparently in either direction. No new computation is required for this check beyond what already powers Figure 1 and Table 1.

## 4.7 Spatial placebo: 5–20 km donut as faux donors (RC7)

**Concern.** The 20 km donor buffer is meant to keep donors fully outside any Costco’s competitive zone, but this assumption is never verified in the headline analysis. The 5–20 km donut (stations currently excluded from both treated and donor groups) is a natural laboratory: treat the donut as a faux treated unit and check whether synthetic-control spuriously finds a ‘Costco effect’ there.

Two outcomes are both informative. (a) The donut shows an effect: Costco’s competitive footprint extends beyond 5 km and the headline estimate is conservative (the true effect is probably larger). (b) The donut shows no effect: the Costco impact is concentrated within 5 km and the 20 km buffer is more than sufficient.

**Interpretation.** The donut’s post-treatment gap is the diagnostic. A statistically clean zero (gap small, CI straddling zero, MSPE ratio near 1) means the headline 5 km treated definition captures the Costco footprint without leakage. A clearly negative donut gap means the footprint extends past 5 km and the headline is conservative (true effect is probably larger). A clearly positive donut gap would suggest substitution toward donut stations, which would be unusual but worth flagging.

Table 16: RC7 spatial placebo. The headline 5 km ring and the 5-20 km donut share the same >20 km donor pool. If the donut shows a Costco-style negative gap, the headline 5 km estimate is conservative (the effect extends beyond 5 km); if the donut gap is near zero with a CI that includes zero, the headline is contained within the 5 km footprint and the 20 km buffer is safely conservative.

| Specification        | Costco         | Pre mean (¢/L) | Post gap (¢/L) | % red. | MSPE ratio | 95% CI         |
|----------------------|----------------|----------------|----------------|--------|------------|----------------|
| 5 km ring (headline) | Casuarina      | 143.04         | -0.91          | -0.63  | 1.00       | [-0.91, -0.90] |
| 5 km ring (headline) | Coomera        | 151.27         | -0.60          | -0.40  | 2.08       | [-0.56, 0.85]  |
| 5 km ring (headline) | Lake Macquarie | 140.23         | 1.52           | 1.08   | 1.40       | [0.45, 1.49]   |
| 5 km ring (headline) | Perth Airport  | 140.74         | -3.40          | -2.41  | 4.31       | [-3.35, -1.37] |
| 5-20 km donut (RC7)  | Casuarina      | 143.38         | 0.28           | 0.19   | 1.30       | [-0.20, 0.27]  |
| 5-20 km donut (RC7)  | Coomera        | 150.35         | 1.63           | 1.09   | 2.63       | [0.83, 1.61]   |
| 5-20 km donut (RC7)  | Lake Macquarie | 134.18         | -3.66          | -2.73  | 2.24       | [-3.56, 0.27]  |
| 5-20 km donut (RC7)  | Perth Airport  | 139.96         | -0.01          | -0.01  | 0.57       | [-0.40, -0.02] |

## 4.8 Summary

**Bottom-line read across the checks.** RC1 finds non-zero placebo gaps for all four Costcos. Perth Airport (-1.96 ¢/L, CI [-2.09, -0.35]) and Lake Macquarie (-3.43, CI [-3.35, -0.24]) have placebo CIs excluding zero, indicating pre-period drift in their donor pools. RC2 flips Coomera positive at 3 km, confirming the Coomera headline as inconclusive; Perth Airport stays negative at all three geometries. RC3 rescues Casuarina at 10 km (-4.49 ¢/L, MSPE 3.19) once the sparse 5 km ring is widened. RC4 shows Perth Airport at -3.65 ¢/L with COVID dropped, slightly larger than the headline. RC5’s holdout RMSPE ratios (1.22–2.03) indicate the synthetics generalise out-of-sample. RC6 (CIs) is reported throughout §3. RC7’s Lake Macquarie donut at -3.66 ¢/L versus the 5 km +1.52 confirms the wrong-sign headline as a ring artifact.

A Costco-level estimate weakens whenever a check inverts the sign of its gap or pushes the post/pre tracking-error ratio below roughly 3. The §5 recommendation discounts Coomera (RC2 sign flip), reads Casuarina through RC3, and treats Lake Macquarie’s positive headline as a 5 km artifact via RC7. Perth Airport’s gap survives the checks, with the caveat that RC1’s WA donor-pool drift suggests some portion of the headline could reflect that drift, so the Costco-specific component is plausibly somewhat smaller than the headline -3.4 ¢/L.

## 5 Recommendation

### 5.1 What the data actually shows

The four per-Costco point estimates tell different stories, and the aggregate mean (-0.85 ¢/L across the four) obscures more than it reveals.

- **Perth Airport (-3.4 ¢/L, MSPE ratio 4.31, 95% CI [-3.35, -1.37]).** The cleanest case in the panel. It is the only Costco that clears the standard post/pre tracking-error threshold ( $\geq 3$ ) and the only one whose donor-permutation CI sits squarely below zero. RC4 confirms the result is not driven by the COVID demand collapse: dropping Mar–Dec 2020 from both panels moves the gap from -3.4 to about -3.65 ¢/L. One caveat: RC1’s placebo-in-time gap for Perth Airport is -1.96 ¢/L with a 95% CI excluding zero, indicating some pre-period downward drift in the WA donor pool. The Costco-specific component is plausibly somewhat smaller than the headline -3.4.
- **Casuarina (-0.91 ¢/L at the headline 5 km radius; -4.49 ¢/L at the RC3 10 km radius, MSPE 3.19).** The 5 km ring contains only ~3 contributing stations on average and buries the signal in single-station noise; widening to 10 km exposes a substantially larger negative gap that meets the MSPE bar. We take RC3 as the credible read of Casuarina; the headline 5 km row is too sparse to trust.
- **Coomera (-0.6 ¢/L, MSPE 2.08, CI [-0.56, 0.85]).** The point estimate is small and the CI straddles zero; RC2’s alt-radius sensitivity (the 3 km / 15 km variant flips Coomera positive) shows the result is not robust to the geometry choice. Inconclusive.
- **Lake Macquarie (1.52 ¢/L).** The wrong sign at the headline 5 km radius. RC7’s spatial-placebo donut for Lake Macquarie, by contrast, comes in materially negative. This suggests the 5 km ring is the anomaly (likely one or two stations dragging the mean up), while the donut behaves as expected. We read Lake Macquarie’s positive headline gap as evidence that the 5 km treated definition is fragile when only a handful of stations feed it; it doesn’t undermine the Costco mechanism.

In short: the *Costco-lowers-competitor-prices* story is real, but it is clear at Perth Airport, plausible at Casuarina once the ring is widened, and inconclusive at Coomera and Lake Macquarie given their fragility to geometry and station-count.

### 5.2 What the ACCC should do

Three recommendations, each scoped to what the evidence supports.

1. **Treat low-cost discount entry into a concentrated local fuel market as pro-competitive in merger and acquisition review, but require a market-by-market test rather than a blanket presumption.** The headline four-Costco mean overstates how broadly the effect generalises; the strongest case (Perth Airport) is also the smallest pre-period and the one closest to the COVID confound.
2. **Publish the *local* (sub-state, sub-postcode) competition map** that this study uses (5 km treated ring + 20 km donor buffer) alongside the existing ACCC fuel-monitoring dashboard. The state-median series the ACCC’s public reports lead with average over too wide a footprint to surface effects of the magnitude documented here. Where station density is low (Casuarina-style rings), the dashboard should also report a 10 km radius. RC3 shows that a narrow definition can bury a real signal in noise.
3. **Re-evaluate the question in two to three years.** Auburn (NSW, opened Aug 2025) and Canberra Airport (ACT, opened Nov 2023 once the ACT joined NSW FuelCheck) will both have  $\geq 24$  post-treatment months by then. Two more clean test cases (bringing the panel to six) would materially strengthen or weaken the policy claim.



### 5.3 Back-of-envelope consumer impact

Honest BotE: we cannot defensibly multiply a -0.85 ¢/L four-Costco mean by ten Costcos and a ring volume, because three of those four estimates are weak. Instead, we anchor the BotE on Perth Airport’s clean result and show what the implied savings would be *if* that point estimate generalised to other Costco rings.

**Inputs (shared across both scenarios).**

- Ring population (assumption): 100,000 residents inside a 5 km radius, a mid-density urban catchment. Casuarina and Coomera sit somewhat below this; Lake Macquarie and the WA metro Costcos sit at or above.
- Petrol consumption per capita (ABS Australian Petroleum Statistics): ~900 L/year.
- Competitor share of ring volume: 70% (one Costco competing against roughly 20 incumbent stations).
- Annual competitor volume in one ring: ~63 ML.

**Scenario A: Perth Airport case (3.4 ¢/L price reduction).** Savings of about **A\$2.14 million** per ring per year. Across the four treated Costcos, this would amount to roughly A\$8.6 million *if* Perth Airport’s magnitude generalised. At ten Costco rings nationwide, A\$21 million / year.

**Scenario B: four-Costco mean (0.85 ¢/L price reduction).** Savings of about **A\$0.53 million** per ring per year. Extrapolated to ten Costco rings, ~A\$5 million / year.

Both numbers are sensitive to all three assumptions. The reason the two scenarios are so far apart is the Lake Macquarie / Coomera drag on the four-Costco mean; the per-Costco picture in §5.1 is what matters for policy here, since the four-Costco mean is dragged toward zero by two fragile estimates. We report both so the ACCC sees the full range, but the **policy-relevant figure is closer to Scenario A’s per-ring magnitude**, applied only to Costcos with station-density profiles that support the 5 km treated definition.

### 5.4 What this is *not*

The analysis estimates the *competitor* response to Costco entry. It does not estimate: Costco’s own pricing; consumer welfare more broadly (which would include time costs, switching costs, and Costco’s membership fee); or entry by other discount brands. The recommendation should be read in that scope.

## 6 Limitations

Five issues sit outside what §4 can resolve. They bound the scope of the §5 recommendation.

**n = 4 treated Costcos.** The headline mean averages across four units, so individual point estimates carry more weight than the aggregate. Casuarina and Perth Airport (the two most fragile cases per §1’s data-quality notes and §2’s COVID discussion) are 50% of the panel. The pooled mean is intentionally reported alongside per-Costco values in Table 9 so a reader can decide how much to weight each row.

**Casuarina sparsity.** Casuarina’s 5 km ring contains only ~3 contributing stations on average. Its pre-period tracking error in Table 10 is correspondingly higher than the other three Costcos, and its donor-permutation 95% CI is wider. RC3 widens the ring to 10 km as a stability check; if the 5 km and 10 km gaps diverge materially, Casuarina’s per-Costco point estimate should not be read as precision evidence even when the sign is informative.

**Perth Airport / COVID confound.** Treatment falls 19 February 2020, approximately four weeks before Australia’s COVID demand collapse. RC4 re-estimates with March–December 2020 dropped to net out the pandemic, but the residual concern is that the synthetic counterfactual is built from a small WA donor pool (36 postcodes) and the WA-wide pandemic shock may not have been absorbed identically across donor and treated postcodes. We report both specifications.

**Six Costcos excluded.** Marsden Park, Auburn, Canberra Airport, Casula, North Lakes, and Ipswich do not satisfy the  $\geq 24$  pre /  $\geq 12$  post inclusion rule (§1.2). The remaining four Costcos may not be representative. Marsden Park and Casula are pre-FuelCheck NSW entrants, North Lakes and Ipswich predate the QLD scheme, and Canberra Airport opened just before the ACT joined NSW FuelCheck. The headline generalises most cleanly to *future* Costco entries in mature state registry coverage. The recommendation in §5 should be read with that scope.

**Same-state donor pools only.** Each synthetic is built from within-state donor postcodes so state-level shocks (excise rates, refining capacity, regional shipping costs) are absorbed by construction. The cost is a smaller donor pool, especially for QLD (56 donors) and WA (36 donors); WA in particular has limited postcode-level price heterogeneity to lean on. Cross-state donor pools would widen the pool but reintroduce the state-shock confound, and we judge the within-state restriction the right tradeoff for a fuel-market analysis.

**External validity beyond Australia.** Australian retail fuel is price-cap-regulated at the wholesale level (via the Petroleum Products Subsidy Scheme historically and through fuel-excise indexation today) and has thinner retail competition than the US or much of Europe. The headline gap should not be extrapolated to other jurisdictions without rebuilding the analysis on local data.

## Appendix: how this analysis is reproduced

This `.qmd` reads its five input CSVs from a public Dropbox folder, fits synthetic-control models with `tidysynth`, and renders to PDF in a single `quarto render` call. The full reproduction procedure (required R packages, the Dropbox folder URL, locations of upstream pipeline scripts) is documented in `analysis/README.md`.